lecture 20 Polynomial Interpolation: Runge Phenomenon and Advanced Methods @April 15, 2026

Summary

Executive Summary

- The Runge phenomenon shows that increasing interpolation nodes with equally spaced points can increase error, especially near interval endpoints▸
- Chebyshev points provide optimal node placement by clustering nodes at boundaries, minimizing the error product term $prod_{i=0}^{n}(x - x_i)$▸
- Newton's divided difference method enables efficient incremental interpolation when adding new data points, avoiding full recalculation required by Lagrange▸
- Divided differences provide a practical way to bound interpolation error using only data values, without needing to differentiate the unknown function f▸
- Hermite interpolation matches both function values and derivatives at nodes, requiring polynomials of degree at most $2n + 1$ for $n + 1$ nodes▸
- Final exam will cover all material through interpolation; midterm retake offered after final exam with grade being the maximum of both attempts▸

Conceptual Map

Runge Phenomenon

(A) Core Concept & Context

The Runge phenomenon describes the counterintuitive behavior where increasing the number of equally spaced interpolation nodes does not necessarily improve approximation quality, and often dramatically worsens error near interval endpoints.▸ This phenomenon was discovered by Runge while studying the function $f(x) = \frac{1}{1+25x^2}$ on the interval $[-1, 1]$, challenging the assumption that more data always yields better interpolation.▸

(B) Detailed Explanation

The phenomenon has two distinct sources of error. First, the $n+1$ derivative of f (denoted M in the error bound) can grow with n depending on the function.▸ Second, and more critically, the product term $\prod_{i=0}^{n}(x-x_i)$ exhibits large oscillations near the endpoints when nodes are equally spaced, and this error source is independent of the function being interpolated.▸ The error becomes particularly severe as x approaches the boundaries a and b of the interpolation interval. This is not a defect of any particular interpolation method but rather a property of the unique interpolating polynomial itself.▸

(C) Formal Notations (LaTeX)

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Key formulas

- Runge function: $f(x)=\frac{1}{1+25x^2}$ on $[-1,1]$
- Error product term: $\prod_{i=0}^{n}(x-x_i)$
- Interpolation error: $|f(x)-P_n(x)|=\frac{|f^{(n+1)}(\xi)|}{(n+1)!}\,|\prod_{i=0}^{n}(x-x_i)|$

where $\xi\in(a,b)$ depends on x, $P_n(x)$ is the interpolating polynomial of degree n, and M is an upper bound for $|f^{(n+1)}|$

(D) Critical Connections

- Connects to the interpolation error theorem from previous lectures, explaining why the error formula can fail to predict convergence▸
- Relates to Gibbs phenomenon in Fourier series approximation, representing similar boundary oscillation behavior▸
- Motivates the need for non-uniform node spacing strategies like Chebyshev points▸

(E) Evidence from the transcript

"The approximation of the interpolant gets worse as n increases, so as the number of nodes increases, and the error is larger near the endpoint."▸ "The problem comes from the product of x minus x0 all the way to x minus xn."▸ "Things to remember, there are two sources of error if you are interpolating. One comes from the n plus 1 derivative of f. But there's one worse source of error that comes from the fact that you have equally spaced interpolating points."▸

Chebyshev Points

(A) Core Concept & Context

Chebyshev points are optimally placed interpolation nodes that minimize the maximum value of the product $\prod_{i=0}^{n}(x-x_i)$ over the interval, thereby addressing

the function-independent source of Runge phenomenon.▸ These points cluster near the interval boundaries where equally spaced nodes produce the largest errors.▸

(B) Detailed Explanation

Chebyshev points are obtained by solving the optimization problem $min_{x_i} max_{x in [a,b]} left|prod_{i=0}^{n}(x - x_i)right|$.▸ Geometrically, they correspond to equally spaced points on a semicircle that are then projected onto the x-axis via the cosine function.▸ For the interval $[-1, 1]$, the formula yields nodes with natural clustering at the boundaries. For arbitrary intervals $[a, b]$, the points must be rescaled and translated.▸

(C) Formal Notations (LaTeX)

∑ **Key formulas**

- Chebyshev points on $[-1, 1]$: $x_i = \cos\left(\frac{2i+1}{2n+2}\pi\right)$ for $i = 0, 1, \ldots, n$
- Optimization formulation: $\min_{x_i} \max_{x \in [a,b]} \left|\prod_{i=0}^{n}(x - x_i)\right|$

where $a = -1$ and $b = 1$ for the standard case

(D) Critical Connections

- Directly addresses the product term error source identified in Runge phenomenon analysis▸
- Applicable when the user has control over data generation rather than being given fixed data points▸
- The non-uniform spacing principle contrasts with the equally spaced nodes assumption used in earlier error bounds▸

(E) Evidence from the transcript

"He said, okay, so I'm gonna try a set of points. These are called the Chebyshev points."▸ "What you are actually doing is you are calculating your Chebyshev points, which are the points on the x-axis... They are equally spaced nodes in a half a circle. And then because you're taking the cosine, you are projecting them onto the x-axis."▸ "You can see here that the order of the error is much smaller in the case of having Chebyshev points as opposed to having equally spaced points."▸

Newton's Divided Difference Interpolation

(A) Core Concept & Context

Newton's interpolation constructs the unique interpolating polynomial using a basis that enables incremental updates when new data points are added, unlike Lagrange interpolation which requires complete recalculation.▸ The method expresses the polynomial in nested form with coefficients called divided differences that can be computed recursively from function values alone.▸

(B) Detailed Explanation

The Newton form writes $P_n(x) = a_0 + a_1(x - x_0) + a_2(x - x_0)(x - x_1) + cdots + a_nprod_{i=0}^{n-1}(x - x_i)$.▸ Each coefficient a_k is a divided difference computed from previous divided differences. When a new node x_{n+1} is added, only one new coefficient a_{n+1} must be calculated; all previous coefficients remain unchanged.▸ The divided differences are computed using a table structure, though the computational complexity is $O(\frac{3n^2}{2})$ for construction versus $O(n^2)$ for Lagrange.▸

(C) Formal Notations (LaTeX)

∑ **Key formulas**

- Newton form: $P_n(x) = a_0 + a_1(x - x_0) + a_2(x - x_0)(x - x_1) + \cdots + a_n \prod_{i=0}^{n-1}(x - x_i)$
- Zeroth divided difference: $f[x_i] = f(x_i)$
- First divided difference: $f[x_i, x_{i+1}] = \frac{f[x_{i+1}]-f[x_i]}{x_{i+1}-x_i}$
- Second divided difference: $f[x_0, x_1, x_2] = \frac{f[x_1,x_2]-f[x_0,x_1]}{x_2-x_0}$
- General: $a_k = f[x_0, x_1, \ldots, x_k]$

where square brackets denote divided differences (rectangular parentheses in notation)

(D) Critical Connections

- Produces the same unique interpolating polynomial as Lagrange but with different construction method▸
- The divided differences approximate derivatives: first divided differences resemble slopes, second divided differences resemble second derivatives▸
- Connects to Taylor series through the relationship between divided differences and derivatives (Newton's theorem)▸

(E) Evidence from the transcript

"Every time you add one more point, you just add one extra term, and you just recalculate it, but the a1, the a0, and the a1, and the a2 are going to be the same."▸ "If you're doing Lagrange, you have to recalculate all the terms. If you're doing Newtons... you just add to what you already have."▸ "By construction... if you plug in x0 into the formula, all the terms have x minus x naught. They will go to 0 when x is equal to x0 except for the first one. So your a naught is already f of x naught."▸

Divided Differences and Error Bounds

(A) Core Concept & Context

Newton's theorem establishes that divided differences are equivalent to scaled derivatives, enabling practical error bound calculations using only data values without requiring explicit knowledge or differentiation of the function f.▸ This resolves a fundamental limitation of the interpolation error formula.▸

(B) Detailed Explanation

The interpolation error formula contains the term $frac{f^{(n+1)}(xi)}{(n+1)!}$, which is typically impossible to compute because the function f is unknown or the point ξ is indeterminate.▸ Newton discovered that this derivative term equals the divided difference $f[x_0, x_1, \ldots, x_n, x]$ evaluated at an appropriate point.▸ Since divided differences can be calculated from the data points alone using the recursive table method, this provides a computable bound on the interpolation error.▸

(C) Formal Notations (LaTeX)

∑ **Key formulas**

- Newton's theorem: If $f in C^n[a,b]$, then there exists $\psi \in (a, b)$ such that

$f[x_0, x_1, \ldots, x_n] = \frac{f^{(n)}(\psi)}{n!}$

- Error formula with divided differences: $|f(x) - P_n(x)| = |f[x_0, \ldots, x_n, x]|\,|\prod_{i=0}^{n}(x - x_i)|$

where the divided difference depends on x through $\psi(x)$

(D) Critical Connections

- Generalizes Rolle's theorem to higher-order derivatives and multiple points▸
- Enables practical error estimation when the function f is unknown, addressing the limitation mentioned in the interpolation error theorem▸

- The recursive structure of divided differences mirrors the derivative structure▸

(E) Evidence from the transcript

"There's a way that you can calculate bounds on the nth derivative... from the data itself. Tell me that this is not amazing."▸ "To calculate the dividing differences, I only need to know the data that was given to me."▸ "This is the importance of Newton's interpolating polynomial... now I have a way to calculate bound on this, just based on data."▸

Hermite Interpolation

(A) Core Concept & Context

Hermite interpolation extends polynomial interpolation by matching not only function values but also derivative values at the nodes, ensuring the interpolant has the same "shape" (first-order behavior) as the function at each node.▸ This requires polynomials of degree at most $2n+1$ for $n+1$ nodes, double the degree of standard interpolation.▸

(B) Detailed Explanation

Given $n+1$ nodes, standard interpolation imposes $n+1$ conditions ($P(x_i) = f(x_i)$) and requires $n+1$ parameters.▸ Hermite interpolation imposes $2n+2$ conditions: $H(x_i) = f(x_i)$ and $H'(x_i) = f'(x_i)$ for all nodes, requiring $2n+2$ parameters and thus a polynomial of degree $2n+1$.▸ The construction uses two families of basis functions built from Lagrange polynomials and their derivatives: one family for the function values and another for the derivative values.▸

(C) Formal Notations (LaTeX)

∑ **Key formulas**

- Hermite polynomial: $H_{2n+1}(x) = \sum_{k=0}^{n} f(x_k)\bar{H}_k(x) + \sum_{k=0}^{n} f'(x_k)\hat{H}_k(x)$
- Conditions: $H(x_k) = f(x_k)$ and $H'(x_k) = f'(x_k)$ for $k = 0, 1, \ldots, n$
- Error bound: If $f in C^{2n+2}[a,b]$, then $|f(x) - H_{2n+1}(x)| = \left|\frac{f^{(2n+2)}(\xi)}{(2n+2)!}\right| \left|\prod_{i=0}^{n}(x-x_i)^2\right|$

where $\bar{H}_k(x)$ and $\hat{H}_k(x)$ are basis functions constructed from Lagrange polynomials $L_k(x)$ and their derivatives

(D) Critical Connections

- Extends the Lagrange interpolation framework by incorporating derivative information▸
- Related to the broader context of finding eigenfunctions for differential operators in PDE theory▸
- The degree doubling ($2n+1$ vs n) directly corresponds to the doubling of constraint conditions▸

(E) Evidence from the transcript

"What if instead of just asking for the polynomial to go through the points, I ask for the polynomial to have the same shape to first order as the function."▸ "I have n plus 1 equations of interpolation like I had before... But I also have n plus 1 equations that have to do with the derivatives. So I have 2n plus 2 equations."▸ "If your f, the original function, belongs to C of 2n plus 2... you have a formula for the error."▸

Comparisons and supporting details

Method	Construction Complexity	Evaluation Complexity
Lagrange	$O(n^2)$	$O(5n)$
Newton	$O(\frac{3n^2}{2})$	$O(2
Hermite	—	—

Node Spacing Comparison

Spacing Type	Error at Endpoints	Error in Middle	Use Case	Formula/Description
Equally spaced	Very large for high n (Runge phenomenon)	Moderate	Low-degree polynomials only	$x_i = a + ih$ where $h = \frac{b-a}{n}$
Chebyshev points	Minimized	Optimally balanced	High-degree interpolation, controlled data generation	$x_i = cos

Action Items

- [ ] Study interpolation material through Newton and Hermite methods for final exam on April 29th
- [ ] Practice problem set will be provided (updated version forthcoming)

☐ Optional: Retake midterm exam after final exam to improve grade (final midterm grade will be maximum of both attempts)

Notes

Transcript

you get the grade that you get the grade. But I don't, what I'm saying is I don't give you, I give you less than a grade. Somebody else above does that.

But as long as you have the-B minus? Yeah.

I think you're passing grade. Okay. This is what I do.

Because I'm going to have to sacrifice something for something else. Something has to be.

I understand. Okay. Did you ask about the material? There's a tune that really picks it up. What do you think?

Was it too bad? Okay. I'm sorry I was late.

Why, why? No, I was, there's nothing, so the exam should be on the 29th, the final exam, the 29th of April, unless they change something. I did not request any changes. I don't know anybody that has. Why doesn't this work?

Sorry, oh yes, wrong button.

Sorry.

Yeah, isn't it enforced to have the exam on the 27th of July? Yes, there's...

So the final exams? If there is a final exam, then somebody choose the exam, got the exam dates for you.

I have no reason to request anything different, so it should be 1:00 p.m. on the 29th.

Unless... Somebody asks to change something.

But do we haveMy class is up to.

Yes. The question-What can I tell you? It is what it is. The question is, do we have-my only question is, do we have classes? and the weekOf the original exams? Yeah.

I don't think so. Because you lose a week.

You lose a week, yes. And we've already lost so much. But, uh, okay, but today... One more question.

One last question here. What is the material going to be for the system? Yes. So I was thinking, until after... Up to two days.

Next one's two days. No, no, until the interpellation. So we'll finish the interpellation. which means that the final will not have differentiation or integration or ODEs or PDEs.

So there's going to be root finding methods and then...

So, no, no, no, it's everything.

Yeah, I know. Everything plus nonlinear.

What is new? New, nonlinear interpolation.

Nonlinear? Thank you. Interpolation. That's it. That's it.

Yes. And will we have like a practice program set for updated one? Yes.

And you said everything, meaning there will be problems from the first half of the course as well.

I don't have anything to add. Okay, okay. Okay. Right?

I want to ask simple things.

I can't find it.

Taylor, yeah. So what I was saying before you guys came in was I was thinking for those of you that want, so it's not mandatory, So what we'll have is we'll have a retake of exam one. Sorry, retake of midterm. And then you'll have the final grade for the mid-turn is the max of both of them.

The original.

I was thinking, but again... After, not in class, so I have to do this because I cannot waste hours of class. But after class, so after the final, So will we take the meter after? I don't think you have anything to lose because you have been studying already again. So if you want to upgrade your grade, as I'm guessing some of you might want, Um...

Okay.

So any other questions? Yes.

Can we get like the-The graded midterm papers. Yes, I have it here.

But do you mind if I give it after the class?

Yes, no problem.

Yes, I know exactly what you did. Trust me, if I had you guys... Like, who? Who would think this? What is this? So I do a lot of I think I verbalize a lot when I'm grading. So my kids know that I'm grading because of the things that I say. Okay,

but today, it's an easy day. It's been a good day for me so far. So I'm guessing it can only end well. And I will be looking at the second part of polynomial interpolation.

And Maybe, time allowing it, I'm going to go into piecewise polynomial interpolation. So this is where I'm going to go.

So what we have been doing is If you've noticed last time, I'm looking, I'm given a set of points, and there's one. Okay, and I want to interpolate those points. which means that I'm looking for an approximation for a model. that data feeds the data that you have And the idea is that you want to make sure that the interpolant, the polynomial interpolant, goes through the points. This is what I have been doing so far.

And we had a theorem that expresses the difference between my interpolate, so this is my interpolate, and this is the true function Right? And so if I subtract one from the other, I peek. some sort of absolute value, you see that I have a little formula for the error. So this is the first interpulent error here. And there's two things that I would like to address, maybe three. The first one is that all the points, the data points or the nodes for the interpolation, so the Xi's, They are n plus 1 as I mentioned, but they are distinct nodes in a given interval, a, b.

So you can have x0 equals to a and xn equals to b, for example. So the distinct is important. The other thing is that there is no particular spacing between these two nodes. And the other thing that I would like you to remember is that, of course, you are assuming that your function f has many derivatives. But... So you have a formula for the error. But if you don't have a formula for F, You cannot Calculate the error because you don't have a formula for F. So you cannot differentiate it.

Maybe you are only given the points themselves. But even if you have the formula for F, you don't know what the chi is.

OK, you don't know what this xi is.

But of course, we can bound. If you know f, you can bound the n plus 1 derivative. then you have a way to calculate the error. Okay, so this is what we did last time, assume me, that we have equally spaced nodes. So if I have equally spaced nodes, there's a formula for this product here. bound for that product if you have equally spaced nodes. And this m that you have here is actually an upper bound for the n plus 1 order derivative of the function n.

OK, so this is actually the second interpolation They are a theorem corresponding to equally spaced nodes. But again, the problem is, I was mentioning, so the H is, this H is the difference between the two nodes, like subsequence nodes, and the M is the upper one. The hard part is to get the upper arm. Okay, so please bear in

mind that upper bound is hard to get because most of the time you don't have the F that you can differentiate. All right.

This is going to be important for the next interpolant that we're going to study, which is neutrons. Okay, so in practice, in practice,Um... How do we do? So if I don't have and I cannot differentiate it, how do I work, how do I measure whether or not the error that I have in my interpolate, in my polynomial interpolate is it big or is it small? So there are basically two approaches for this. This is again chapter 10.5 if you want to know more.

From Asher and Reef. But the idea is that the first approach is the first one that you see there is very useful if you know that your F doesn't vary a lot between A and B. So we get incubo. Um... So then what you do is you calculate interpolating polynomials of different orders, so with different nodes. Okay, and you compare them. If they are relatively close, when you plot them, then you're on a good path.

to having a small amount of error Okay, and then the other approach is the approach that we usually take in AI is that you divide your number of nodes, your nodes into the training set and the testing set. and then you compare what's happening. The problem here is that you need to generate more points, more dyadron points, right? So that you can keep the order of the interpellate. So I'm not going to go into the details of this.

Again, as I mentioned, there's a lot of information in Asher and Grief if you want to know more. But those are the two types of things that you can do, like easy things that you can do to try and understand what is the error, to try to understand if your interpolant is good or bad.

Okay, because you don't remember. Again, you don't have... You generally don't have the F to differentiate. So then you can get the M in the second interpolating error. Okay, so this was the, so that's the formula.

So the questions that we had last time, if you remember, I have these four questions, reasonable questions. So the first one was, what is the error involved in approximating a function by an interpolating polynomial?

Can you have me please there? Okay, that's the formula. So that's the error.

Is f, so is the function that you are trying to interpolate well approximated in intermediate points? The formula is still there because that's valid for any x. So that's the upper bound for any x. So that guy, m, multiplied by h power n plus 1 over 4n plus 1 does not depend on the x. So you can just work it and see how much that is, and that's the maximizer for the error that you have. So we still, so we answered one and two. We are going to answer three and four.

So the question is, will more nodes, the two questions, will more nodes increase the quality of the interpolating polynomial?

And the answer is yes, if we're careful. Yes, but. So the but comes after.

Can we choose the location of the nodes to increase the quality of the interpolating polynomial? Yes, and you shoot. So this is what I've been working, not me, but what we work on.

So this is the work of, I don't know how to say this, okay?

So this guy, Rungi was working on with this function. So 1 over 1 plus 25x squared.

And he worked so much that the function actually got his name.

So the function, the domain of this function is in between minus 1 and 1, and the graph of the function is what you see there.

So this would be-the function is 1 at 0. And you know, it's--a very nice function, right?

So it's a rational function. I mean, there's no discontinuities. It's a beautiful function. And he was trying to Interpolate this function. And at that time, he tried different methods Interval length, so with different orders. Okay, so this would be the turbulence of order two here because he had three points. And then he thought at the time, it's actually interesting to go and read what he was like his notes.

And he thought that obviously the more points you had, The more data points you have, then the better the interpulent would be. So the smallest error. But you can see that this is not true. because For this many amount of nodes, you can see that you have 1, 2, 3, 4, 5, 6, 7, and so on. You can see that you have a big error here, especially, so the error is still big here, but so basically what I'm trying to say is the error gets bigger as you get closer and closer to the end point.

So he was, you know, trying to figure out why this is true, but the conclusions were, so this is like a superposition graph of what we had before. The idea is that the approximation of the intercalant gets worse as n increases, so as the number of nodes increases, and the error is larger near the endpoint. So you can see here, right? So this is a measure of the error as you get close to negative 1 and 1.

So he was like trying to understand why, And he was looking at the formulas that he had. So this is called this idea that as you increase And So as you increase the number of points, and hence as you increase the order of the interculent that you are working with, You actually are not getting a smaller error, especially in the endpoints of the interval. is now known as Rundsch Phenomena. And it's kind of close to Gibbs phenomena, if you know that, when you approximate.

free stuff, I don't know if you've seen this before, but it's kind of the same phenomenon.

And the idea is that you have M. Remember the M in the formula for the error is an upper bound. For the n plus 1 through f, We know that the value of the derivative, so this value, changes not only with n, but it also changes with xi, so it also changes with x because xi is a function of the x. So it's a function of where you are trying to measure the error. And we know that if you do the calculations, so if you differentiate the Runge formula or function, so we have like a closed formula, so you can differentiate n plus 1 times, you can see that This Value here grows at the end points.

So this was one reason. So the interpolation error increases with the value of n. And the problem-so this is one problem. that he detected So this problem here depends on the function that you are working with, right? So when you differentiate the function, The derivative depends on the function that you are working with. But, so he did these calculations for this particular function and he did for other functions, But then he realized that for some functions, this was not the case.

Not the Runge function, but for other functions.

So the problem had to be, he thought, independent of the function itself, of f itself. And it turns out that the problem comes from the-so the problem of having the error big at the end points, is not related to the function. It's not only related to the function, I should say, but it's actually related with the product of x minus x0 all the way to x minus xn. So if you Graph. this product So I'm looking at the graph.

of the product of X minus Xi, right?

So if you go to the formula, So if you let me just get my pen, you can see here.

So if you go to the formula So if you go to the formula here, You have here the product of the distances, right? And if you plot it in this particular case, so what I did was I plot here here for n plus 1 equal to 10. So this would be this graph. So what am I plotting? So this here-is the graph of the product from i equals to zero all the way to n. x minus xi. So this graph that you see here is the graph for this function when you are looking at n plus 1 equals to 10 and n plus 1 equals to 16.

And you see this problem here, right? So this is 10. Can you see?

10, if I do this, oh, sorry. So this is 2 to the 10th of minus 3 and this is 2 to the 10th of minus 4.

So we still have a big error as you approach the end point. So this is as you approach minus 1 and 1. Okay, so what I'm trying to say is, The sources of the Runge phenomenon, so the fact that you don't get a better approximation as you increase n, Depends, and the fact that you have a very bad approximation as you

approach the endpoints has two sources. You can have the source coming from the F. So from the n plus 1 derivative, but that's not the only source.

There's a source of error that is independent of the function that you are trying to interpret. And that source has to do with the term, the product of x minus xi. So he realized that this was true, And so he said, "Okay, but what if instead of Trying to work with equally spaced points, why don't I change to unequally spaced points? Okay, so he said, okay, so I'm gonna try a set of points. These are called the Chebyshev points.

They will appear over and over again. I will tell you how to get those. But what he did was he then calculated again the graph of that function. For these Chebyshev points that you can see are not equally spaced. So they are, you have more points closer to the boundary of the interval that you are considering. In this case, minus one, one, so AB. And this is for a hand n plus 1 equals 2 n. And n plus 1 equals to 16.

And you can see here that the order of the error is much smaller in the case of having change of points as opposed to having equally spaced points. So, things to remember, there are two sources of error if you are interpolating. One comes from the n plus 1 derivative of f. But there's one worse source of error that comes from the fact that you have equally spaced interpolating points, so equally spaced nodes.

So he said, Lund did, he said, "Okay, so what if I, instead of using Ah. This point, so equally spaced, what I'm going to do is I'm going to... not use an equidistant distribution of points, So this is like basically like a rule of thumb If you're thinking about interrelation, So I said I have four points. Do not use equidistant points. This is not usually a good idea unless the degree of the polynomial is quite small, so you don't have a lot of oscillations. Because if the degree of the polynomial is small, you don't have space, you don't have the flexibility of having the amount of oscillations.

So you don't see those big oscillations at the beginning and end of the interval.

So usually what we do is we cluster the nodes at the boundary. Okay, I will tell you how. But I wanted to point out, and this is my last point here, is that this is not a problem of the only interpolant that we know, which is the Lagrangian interpolating poly, the polynomial, because remember the interpolating poly is unique. So it's going to be something that you're going to see across Any way that you get to the interpulent, you're gonna have the same problem.

Because the interpill and pulling angle is the same.

So it has nothing to do with the way you used to construct the polynomial. It has to do with the fact that the product of x minus xi is not well behaved at any point. Okay, so how did Grunge thought about it, he said, in his smartness. I'm going to go

and choose my nodes. So the xi's are the nodes. So that what I'm minimizing in this Xi is the maximum value in the interval of the product.

Sorry, you said that depending on, like, by changing this frequency of the points and the endpoints, in the boundaries, the polynomial doesn't change, but don't we fit the polynomial to those x i's?

No, what I mean is...

Going back, so the problem So you're thinking about this?

Yes.

So if you keep, you might have thought, you say, So every time you change the notes, yes, you have a different interpolating polynomial.

What I'm trying to say is that you might think that the problem would be-coming from the way you construct the polynomial. And in this case, we only saw one type of construction, which was Lagrange, right? What I'm trying to say, this is not true. It has nothing to do with the construction because the interpolating polynomial given a set of nodes is the same.

This is what I-sorry. Maybe I should-No, I shouldn't do-write it down. Yes, thank you for the question.

So if you solve that problem, I solved it once. If you solve-so this is an Emax problem.

I'm looking at you because you keep repeating everywhere.

So I've solved this problem. So if you solve this problem, the solution for the Xi is actually has this formula. So it's a cosine of 2i plus 1 over 2n plus 1 pi. Okay, we eye from zero to n, so you have all the nodes that you need. Okay, and this is true Where your a and b are equally respectively to minus 1 and 1. So this is, these guys here are called Chebyshev points. and they are defined in the interval from minus 1 to 1. So your a is equal to minus 1, and your b is equal to 1.

But if you are not working in an interval a, b, what you need to do is, well, you need to rescale.

and translate the note that you got from minus one to one into a B.

So this is not a problem, at least we know how to do. And so this is an optimization problem. that you can actually do. But what I really would like you guys to see is what we're doing. When you create, when you're calculating those cosines, what you are actually doing is you are calculating your, so you're calculating your Chebyshev points, which are the points on the x-axis, so you have them here, x0, this is the minus one, x1, x2, and x3.

What you're doing is They are equally spaced nodes in a half a circle. And then because you're taking the cosine, you are projecting them onto the x-axis.

And those are the change-in points.

So if you do this For a large I, so here I only took I I equals to three, but if you do it for a large I, you're gonna see that the Chebyshev points, they cluster at the boundaries. So you have more refinement in the interpolation at the boundaries. And this is what the Chebyshev points are. So our Chebyshev points are if you're doing an interpolation and you are expecting to have a lot of points, you have to generate your data.

And you are not given the data.

You have to generate the data according to the Chebyshev terms. because otherwise you're not getting a you're going to see the Rundsch phenomena.

And once you see it, you will remember. So if you try to do equally spaced, points, then you are understanding that the error, you just output the error, you see that it's large. When you ask your code to graph the intercalant polynomial, then you will see that you will have a big L, like large oscillations very close to the boundaries, and then you will see that, I mean, you would immediately recognize it as being

And the third term will be A2. So this is what we're looking for, right? Because the nodes we already have. a2 is going to be multiplying x minus x0, which was the basis that I already had before. and then it multiplies x minus x1. And likewise, A3 is going to multiply the previous ones. These were the ones that I had before.

And then I multiply by the extra term, x minus x2.

Until you reach the end. And your term corresponding to the coefficient a is of n is going to multiply everybody before plus this new term. And this is going to be important in the way you construct the point of view. The interpellant boy.

So this is what I would like you guys to understand is that Every time, so this is what I said, every time you have a new note, You just... One more turn, but the previous ones are all the same. So this is what I need. So how do I get this? So this is the formula. Okay, so what I was trying to say is by the way you construct the polynomial, Every time you add one more point, so you have an extra xi, let's say you are doing your experiment in the lab, and at the same time as you get the data, you are recalculating your interpolate.

If you're doing Lagrange, you have to recalculate all the terms. If you're doing new plants, Going back?

Let's say I have this, right? I have only two backup points, x naught and x1. Now in comes the next one. The next, well, in comes the next back of 1 corresponding to

x2. So you're going to increase the order of your intercalant because you have one data point more.

So you just add to what you already have, you just add this u term, and a3 you just recalculate it, but the a1, the a0, and the a1, and the a2 are going to be the same. Okay, so how do I calculate, and this is why we like Newton's construction. because we can easily add more data points to what we already have if you want to increase the order of the interval. So this is one big advantage, I mean, opposition to lag ranges.

What is the not so nice thing So, you know, When you decide, okay, I have a unique interpulent, What method am I going to choose to come up with the interpulant? And then this depends on what you have and depends on what you know. If you are not expecting to add more data points, I would just go with Lagrange because you don't have to do any calculations, right?

you are given the bases, so the live range bases, and you just multiply the corresponding live range base with the height of the point that you have at that particular node.

But here, you know, so the AIs, they have nothing to do with the Fs. I mean, they have, but it's not obvious. The advantage is you just add one extra term. So all constructions have their own advantages.

And the way you choose the construction actually depends on what your goal with the interpolant is.

Is this clear what I'm trying to say? No, yes.

Yes?

Because the equation depends on the data itself. That's why we have to recouple it if we want to add more. Or lab rats.

For Lagrange, the bases depend on the points that you have from the beginning.

So if you have one extra one, When I say when you have one extra data point, you have to recalculate the basis. Because remember, the basis were x minus x1 multiplied by x minus x2 and so on.

The advantage of Lagrangian, if I go here, Maybe I should, if you go to the slides before, Maybe I can go. Let me see. Do I have it here? Yes. Oh, not here. Where am I? I don't know. This is not what I want. I should be here. Okay, am I here? No, notes. Oh, here. So this is what we were doing last time. Oh, here, lagrange, lagrange, lagrange, lagrange, lagrange, lagrange.

Labyrinthial branch, labyrinthial branch.

Here, Lagrange, these are each one of the L's. You see the L's, they depend on all the points. So every time you have a new point, you have to recalculate each one of the lag ranges. But the advantage is Once you recalculate the L's, you know exactly that you need to multiply each one of the Lagrange. polynomials by the height. Remember, because these 14 and 24, 25 and so on are actually the y's of the points that you are given.

So you know you don't have to calculate anything other than the Lagrangians, the polynomials. Whereas here, In Newtons,Where am I? Here, I think. Here Here. Let's say you have two points. So you have x0 and x1. So you can go ahead and calculate the interculant all the way to this pose.

If you have an extra point, so in comes an x2.

You just need to recalculate A3. So you don't have to recalculate everything from the start. This is one big advantage. OK.

So how do I calculate the ace?

Beautiful, beautiful.

So this Oh, this is so nice. Newton, you know. that you can never go wrong with Newton because he always had these beautiful ideas. So this is just amazing. It's like Taylor's. I don't know if you had a chance to look at how Taylor came up with Taylor's expansion. It's also beautiful. And it's so simple. And you go, how come I didn't think of it? Well, because you're not tailors, right? This is the thing.

So this is the polynomial that I want to come up with.

So if I want to calculate this polynomial, I have to come up with a 0, a 1, all the way to a m. By construction, and this is where I think things get cool, If you If you plug in x0 into the formula, All the terms haveX minus X naught. They will go to 0 when x is equal to x0 except for the first one. So your A naught is already F of X naught. Okay, so this is easy. If I'm thinking about, okay, so how do I get a one?

So now I know a 0, right? So this is null. Okay, so this is the first step. Second step. I go and say I'm going to calculate a 1. So I'm going to use the fact that I know that theHe goes through The same point as fx1. So I just say, I just say,That egg. P.M. And x1 is going to be equal to f of x1 because this is the interpolate, right? So if I plug f of x1-I plug X1, I'm going to get This equals to this, that I can solve for a1.

Was this clear? No? No? Yeah? It's a little, like, smooth.

Let's go grab him. Yes, so one important thing, oh my God, he was so smart. So, and you are too, because I didn't see it when I first saw it. So, I mean, I know it has something to do with the gradient, but I didn't recognize it at first. I just thought, oh, this is art. formulas, I'm not going to worry, but you saw it, and he saw it as well.

But just put this aside. So what your colleague is saying that this is a lot like a gradient, I'm sorry, a lot like a derivative, a lot like a slope. Right?

Because it's delta F. over and down the x. But the point is, while I'm doing these calculations, I just need to solve this formula for a1. And I got-The next one. If I want to do A2, I follow the same thing, right?

I just pluck.

I just say that at x equals to x2, now remember, now I know a1 and I know a0. So I'm going to go for a2. So I'm going to plug here, x equals to x2. All the terms that have x2 are going to go away. and which are all the ones that are three or four and so forth. So the one terms that I don't know because I'm looking for these parameters. And I just keep on doing the same thing. So new things Realize that the only things that you have to calculate were what he called the divided differences.

Okay. Hi. So the first divided difference is the 0 divided difference corresponding to a0.

So this guy here This here is a naught. So this corresponds to the first zero divided difference. So a0 is equal to f of x0, which he called-the first divided difference. And the way you notice that you are working with divided differences is because the-Parentheses get to be rectangular, not like round brackets. Okay, so this here, would correspond to the first divided difference X naught X1. So this would be the first divided difference.

So he takes The zero-Livagy difference minus another 0 divided difference. So I have one 0 divided difference. for each one of the notes and divide by the note. So if you do the same thing for So now I have here, I have A naught. Here I have A1, and then I'm going to search for A2. Now, if you search for A2, the formula is going to be-now, to get here, It's not as obvious.

It's nothing like fancy. It's just like arithmetic, but it takes a while to get here.

So you can do it once or twice.

But it's not as obvious as what you have here. But the fact is, you're going to see a pattern. So, and because I remember the pattern, this is how I'm going to write it too.

So A2 is the divided difference between X2 and X0, so between the third and the first point. And here I'm going to have a subtraction. And here I'm going to have a divided difference between x2 and x1. And here between x1 and x0. And here I'm going to have f of x2 minus f of x1. And here I'm going to have f of x1 minus f of x0. So you see the pattern repeats itself. So it's a Delta?

First, store the derivative, if you will, delta x.

So on the topic that I'm always calculating approximations to derivatives. So if here I'm calculating the approximation to a first derivative, as you mentioned before. Here would be a first derivative. Here is a first derivative. So this would be something that is a derivative a delta f prime if you will divided by a certain amount of Delta X.

Okay, and this is, so noticing this pattern, He, the, the, he Newton. He defined all these derivatives. Okay, so, Sorry, did he define all these divided differences? And then, so this comes after, he noticed that these divided differences, they were related to the derivatives of the function itself, and that was really important, I'm gonna tell you why.

But before, I mean now I don't even know if we do this nowadays. Ah.

But this was something that we did like 10 years ago. We actually calculate This divided differences And here you have the interpolating polynomial in terms of the parameters a's. Whereas here, you already have the interpolating polynomial written in terms of Newton's divided differences. And so the only thing I need to do If I want to come up with Newton's intercalate, it's to be able to calculate these divided differences in a-reasonable way And the reasonable way is the dividing differences table.

Okay. I'm not... Sure. So I've taken this picture from Asher and Grief. This is something that we do once in our lives.

It's just a calculation.

But what I needed to remember is that Two things for mutant detergents. Well, of course you need to know how to come up with it. But this is something that you're going to forget in two years, maybe less. But what I need you guys to remember, and this is the important thing that you need to remember for life, is that there's an advantage to working with mutants in turbulence, is that I don't have to recalculate every time I come up with one extra point.

Because the computer does this for you, right?

I mean, if you just calculate Newton's interval length, But you need to, you need to...

to know that you have a certain advantage in using that interpulent.

And this is the things that you need to kind of These are the things that need to stay with you for the rest of your life.

Well, I don't know for your life, but at least for the next academic years. The other one I'm going to tell you after. But before, I'm just going to calculate. I haven't done this in a while. I'm going to calculate the interculent through 1, 1, 2, 3, and 4, 3.

So I have three points.

I'm going to do it using the bi-difference table. Okay. Because you have to see this once, I think. I don't know, did you ever do this for your numerical?

Yeah, so-But the table was a little bit more-organized the National Reef.

This is not, this is not organized?

It was a little bit more, I actually pulled it up. I was interested while you were talking about mutants. You were looking at what you had? Yeah, and I looked it up and I'm looking at it now and it looks just better than... Oh, I'm sorry. Just a little bit. I just called you this one for a match. Just a little bit, it's not much. Um.

Uh, hi. Actually, the only thing I remember I will tell you what I remember from UTEMS.

The things that I was trying to say. I remember that I don't have to recalculate and I remember that it's given information about the derivatives and these are the only two things, the derivatives.

So the divided differences, are actually information with just only the dividing differences. So this is what I wanted you guys to remember, but I can easily calculate if you help me. We can easily calculate this guy. Where is it? Oh, here. Ciao. So I have three points. So this here is my x naught. This is my x1. And this is my x2. So if I do the table, I'm going to copy the table. So my eye, you can see it, yeah. My eye have here X-eye.

I have 0, 1, and 2. 0 and a 2. My points are... One, two, four. And then my f of xi, this is, well, I'm writing it in terms of f of xi, but this is the zero order divided difference. This is f of xi. f of x, 0. This is the first 0 or the divided difference. And this is just one, and then I have three, and then I have three. So this guy here is F. of X1 as we usually call it, but in terms of Newton's, it's divided differences is F of X1 because it's the, Second, zero-order divided difference.

And then I'm going to calculate here xi minus 1 xi. So I don't calculate, oops, I don't calculate the first, I only calculate, Thanks. But the point is, I am just looking at my p. and my P, my interpulent order two is going to be F of X naught. So this guy here And then I'm going to have the second or the first divided difference here. multiplied by x minus x naught. So I need to calculate this one.

And this one comes from this part in the table Okay, let me see if I can, maybe I should have a bigger table. So this is gonna be if, of x1 minus f. of x0 divided by x1 minus x0. So... Just calculate it. So this is going to be... f of x1 is 3 minus 1, which is f of x0 over-x1 which is 2 minus 1 so this is going to be 2 over 1 so this is going to be 2 So this guy here is a two. So I already have everything.

So I only need to calculate f of x0, x1, x2. So I'm just using This is the formula here. On the table. But to calculate this guy, okay, I'm going to use these two guys. So I'm going to calculate the First order divided difference Here, x2 minus x1. This is x1 here. So this is f of x2 is 3-oh, 3 minus 3 is 0, whatever here. So this is 0. So the last one that I need to calculate is the second order divided by the difference.

So it's going to be xi minus 2, xi minus 1, xi. And I only need to calculate it here to multiply by x minus x0, x minus x1. And this is going to be Ah, what I need. So this is gonna be... Zero. minus two. divided by 4 minus 1. So this is minus 2/3, I think, right? So In the end, so the numbers that I need are-Do one here, so one plus. This 2 multiplied by x minus-what is it? 1? Plus, or minus I should say, plus minus 2/3 corresponding to this first order or second order divided differences.

x minus 1, x minus 2. Ta-da! Time to end this video. Okay.

But you see, I mean, you do one, you do two or three.

I mean, you do one or two and then you're done.

Because you understand the topic, right? And now we don't do this by hand, right? We just ask.

Okay, but I've done one, I'm happy. And now...

Okay, in terms of... Complexity? Um...

So construction.

What was it?

So if this is Newton's and this is Lagrange, Or maybe the opposite? So if this is like ranch and this is like a house Mutants is O of 3n square over 2. And then, Lagrangian is O of n square. But then, Mutants is O2m, whereas Ligrand was 5m, if I'm not mistaken. So this is just a comparison between The complexity of construction and evaluation have two different ways of coming up with the interpolated.

Okay Ah, finally. The nice thing, so this is what I was trying to say. Um... The connection between dividing differences and derivatives that I've written down here in terms of a theorem. is nothing more, this is very simple to calculate, you can just read it in Usher and Reif, This is only like an upscaled version of Rolle's theorem.

So there is nothing special, there is nothing very mathematical about this. But you have here the theorem. So, and this is something that you need to remember, I think. because this is really, really relevant. And I will tell you why it's relevant. Maybe you can guess it, but I'm going to say what it is. So let f be a continuous

function with n continuous derivatives in a, b. And then you have x0 all the way to xn.

n plus 1 distinct numbers or points between a and b.

I don't have the time. Oh, OK.

Then, There exists a psi in a, b such that the divided difference on the left The divided difference on the left Corresponding to these many points, is equal to the nth derivative of f at x i over n factorial. So... Why would this be important in our lives? Remember I'm going to go back to the beginning of the lecture.

When I wrote the first and secondUh, Interpolation error zero.

One of the problems that we have If you are calculating the error, responding to this thing here, is that Many times you have the data But you don't have F. You don't know where the data comes from. So you cannot find a bound, you cannot easily find a bound For the n plus 1 derivative, because I don't know what it is. I cannot write it down. But Newton in his wiseness, he said, well, I know. How to calculate it?

I know. Where is it? Here, here, here.

I know how To calculate this ratio, which is the ratio that I had before in the interpolation error theorem, Using not derivatives, by using dividing differences. But if I go back to my Apparently not so good table of dividing differences.

It's the book's table, not yours. Exactly.

You can see that all these, this is the guy that I need. I can calculate this guy using previous-Dividing differences, and the only thing they need is the value of the function at the point. So they only need... The data points. So they only need this stuff here, all these two colors. So there's a way. that you can calculate bounds on the nth derivative So there's a way to calculate bounds on the errors of the interpolation.

Knocked from the derivatives but from the data itself Tell me that this is not amazing. Come on. Only these guys can do this. So this is the importance of Mutants interpolating polynomial. It's not about the polynomial. Nobody cares about the polynomial unless you're getting new data, then you can. But what's really important is that now, whoops, I have a way. Calculate. Bound on this, just based on this cut.

And to calculate the dividing differences, I only need... to know the data that was given to me. So...

If I rewrite The polynomial interpolating error Okay, at those points, this is the formula that I have.

And this is the formula without the derivatives using the previous theorem from Newton's theorem.

And this is, I think,A great contribution to what he did.

Of course, remember this guy depends on the side. So this guy here also, so this guy depends on the point x through psi. Remember it was psi with x.

So now the divided difference depends on the x itself. So if you bound, you have a way. This one we know how to bound for equally spaced-remember, it was 1/4 of something, right?

This one we know. Because now we can write this in terms of the values of x and f.

And you can bound it. Okay.

So two important things to remember about Using utens? Again, if you have more data, this is the guide to use.

And then the second part is you can use experience to actually get a bound on the derivatives.

Derivatives divided by n plus 1. I Okay, but then, We can do better. So what you're going to see now isSo there's a whole, It's like a whole population of interpellers. So we saw a large range,Can you do this?

I'm going to show you one more. This is Hermite.

And You might ask, I did this question, I posed this question, who thinks of these things? Right? How do they... How, I mean, do people just go one day, oh, let me interpolate this way? No, they need the interpellation. So the story is all these people, They were trying to solve partial differential equations. And When you're solving partial differential equations, One good thing to know, so you have like the Derivatives?

And those derivatives correspond to an operator, a differential operator because it's an operator. So it's something that,Ahem.

is applied on functions.

And because it has derivatives, we call it the differential operator. And to solve this PDE, so to solve for the function that solves the PDE, it's very useful to find a basis for those functions. And depending on the operator and depending on the PDE and where you're solving the PDE, You might want to calculate the eigenfunctions of the differential operator. So that those Eigen functions, they function as a basis to define a family of all the functions that are solutions to the GD.

And this is, these things that we are seeing here are, so this inter, this basis for the intercalants are actually eigenfunctions corresponding to some PD.

So this is how They came back with us. So Hermite was solving something. I don't really know what he was solving, but he said, okay, so before, if I have a polynomial interpolator, right?

I am I am asking.

For just one thing, I want my interpulent, so my P, my polynomial, to have the same same value as the function at the nodes. So we write this P of xk equals to f of xk. But what if I ask more? What if instead of just asking for Lee polynomial to go through the points, Bye, Moon! I ask for the polynomial to have the same shape to first order.

as the function.

So he asks not only So if you're doing Haramay polynomials, instead of calling them P, you call them capital H. So instead of asking for the polynomial to be turbulent, You ask also at the nodes for the derivative of the polynomial. to be equal to the derivative of the function at all the points. So you have N plus 1 conditions, assuming that I have N plus 1 nodes coming from the intercalation. And I have An extra n plus 1 equations coming from the fact that you are requesting that the shape, remember that this guy here, the derivative gives you kind of a shape, right?

The shape of the graph of the function at the point. And you are asking for these two shapes to be the same again at the nodes. And this is what a Hermite interpulent is. So Its interpolation is the nodes and the rivets.

If you do this, Then... Okay, so this is 2BZ. But it's not telling you a lot. I know it looks not good, but I have to put it in the same slide because otherwise you lose... So this is telling me What is? The He's telling me two things, or three things. It's telling me what is the ermite polynomial, it's telling me how to construct it, and it's telling me the error. So only three things. I know they don't look that good.

So assuming that f is a C1 function in AB, And there are x0 all the way to xn. Notes, distinct notes.

You may be.

There's a unique polynomial of B of-Degree The unique polynomial of least the green, green with F and F prime at the nodes.

So exactly what I was saying. So it's interpulent and the derivatives are the same at the nodes.

is the Hermite polynomial of degree at most 2n plus 1. given by this formula. So why?

Do I have a degree at most two m plus one. If you gave me n plus 1 notes, Right? Because now I have n plus-when I was doing just interpolation, I had a polynomial

of degree n. Or if you gave me n plus 1, Equations, the interpolation.

Right?

So I have that the p I have, if I had n plus 1 nodes, I had n plus 1 of these conditions. And I got... N plus 1 parameters.

Let me write this because this makes sense. So if you're doing interpolation, you have n plus 1 nodes. So you have n plus 1 equations. that are P of xk equals to f of xk.

And then with this n plus 1 equations, you can determine n plus 1 parameters.

And then when you go and write your polynomial, you write this a1 plus a, sorry, a naught plus a1 plus a2. whatever you want to put here plus A.M. Uh, eh. I don't know, exam. So this, I should write here basis. So base is 1 all the way to a function basis of n. So... You have the basis, you have the constants, the parameters that you need. How many parameters do I have? n plus 1. And I can solve for these parameters using n plus 1 equations.

Does that make sense? This was when I was doing interpellation. Now, so this is interpuration. what we call polynomial interpolation. Now what I have is I have air-mite interpolation.

The guy, Hermite, It also has n+1 nodes.

But he has n plus 1 equations of interpolation like I had before. Oh, let me just call it h. Oh, yes. But I also have n plus 1 equations that have to do with the derivatives.

So I have 2n plus 2. equations.

two equations, so conditions, and with 2n plus two conditions, I can come up with a polynomial that I'm going to call h, because it's the Hermit guy, of 2n plus 1,Perimeters. So I would, no, sorry, 2n plus 2 parameters corresponding, one of the parameters corresponds to multiplying by 1, right? So the last order that you can see in this, it's a polynomial of 2n plus 1.

This is what I mean. because I have 2n plus 2 parameters.

that I can solve here. And this is Guy. is this one that I mentioned here.

So how do I construct this?

So there's two parts: a part that is related to the value of the function multiplying by this basis that you see here and a part that corresponds to the knowledge of the derivatives at those points at the nodes multiplying by the corresponding basis so you have two sets of You can divide the basis in two parts. This is the first part corresponding to f. And this is the second part with coefficients f prime.

So this is very close to Lagrange. So you cannot come with this as a surprise that The basis Both H and H hat are going to be built from Black Ranch polynomials and its derivatives. So...

This here So this is the formula. that you're going to use to construct your might. And this is the formula that you're going to use to construct each one of the bases. And it depends on Lagrange polynomials, Lagrange polynomials here and its derivatives. Okay, so this is one thing.

So this is how to construct the polynomial. A second part of the theorem says that if your f, the original function, belongs to C of 2n plus 2. So you have-And plus 2n plus 2 derivatives that are continuous in the interval of interest. You have a formula for the error, right?

So this guy minus the interpulent with absolute value is equal to this in absolute value.

And again, you have the... n squared plus 2.

as I have before. But now we know how to deal with this, so it's not a longer problem from. Okay, so... Going back, I'm looking at the time, but I'll let people see. Um...

So we have We studied three polynomials.

Three interpolating polynomials. Lagranges, way of constructing. Newton's way of constructing and they are going to give me the same interpolating polynomial if I have the same nodes. Okay.

The other type of polynomial that is interpolating polynomial is more than just interpolating polynomial because it interpolates the polynomial and also has the same shape at the nose to first order because I'm only worried about the first order.

Clear? So this is what we have been doing. These last guys are the air mind blocks.

I've already told you there's like I don't know if you've ever studied this, but when I did numerical analysis on numerical methods, I say all of these guys, and I'll get a little gentle and change their point of view. The point is, this is something that I did not tell you is, when you are trying to find The solutions of the PDEs You want the solutions? to be well behaved in different senses depending on the PDE that you have.

So when I say, Well, behave is that, for example, you can integrate them. You can integrate the square of them. They belong to different vector spaces where the PDEs make sense. So these Hermite polynomials are useful when you are trying to find solutions in the full R, so from minus infinity to plus infinity, because the

Hermite polynomials, as I defined them in the previous slide, you can integrate them between minus infinity and plus infinity.

Right now you don't have to know this, right? I'm only telling you that this interpulence or this polynomial interpolation, it comes from the necessity or appeared first from the necessity to calculate functions of PDEs that you could integrate either in AB or that you could integrate in minus infinity and plus infinity. And this is where these guysHi. are different So I think in P, I'm not sure if in PDE's, We still study this.

Do we ever remember?

At some point we used to teach that we used to calculate functions for different differential operators. And this is where all these guys come into place. Okay, so this is what I wanted to say. Next sign. Before I move on.

We're going to see that, okay, so it looks good, but can we do better? And yes, we can do better.

The idea is that instead of interpolating all at once, so interpolating in one interval, what I'm gonna do is I'm gonna divide that interval into pieces, and I'm gonna interpolate in each one of the pieces. And this is where splines come along. Okay, but this would be... Next Sunday. And now I have a surprise for you, which are your exams.

I have possible solutions. And done. On the whiteboard, I think it's lecture 18, if I'm not mistaken.